A Journal-Agnostic Template for Drafting and Preprint Publication

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Draft compiled 2026-08-01 • v0.1 • 1275 words

Abstract Replace this paragraph with the abstract. One sentence per line in the source, following the vault LaTeX conventions. The abstract is set at full text width in both the one-column and the two-column model, so it reads the same way in either mode.

Keywords Quantum Systems Engineering, Systems Engineering, Quantum Computing

Table 1: Tables use `booktabs` rules.

Process	Share
System realisation	96%
Requirements definition	0.06%

1 Introduction [73 words]

Replace this file with the introduction. Write one sentence per line so that a single changed sentence shows as a single changed line in the git diff.

To remove a sentence, comment it out with a leading percent sign rather than deleting it.

1.1 Motivation

Subsections are available down to three levels. Citations use the numeric style, for example [1].

1.2 Contribution

State the contribution as a short list.

- First contribution.
- Second contribution.

2 Background [45 words]

Replace this file with the background. This stub exists to demonstrate that displayed mathematics, figures, and tables all carry line numbers correctly in draft mode.

A displayed equation stays inside the line numbering.

$$I(A;B) = H(A) + H(B) - H(A,B) \tag{1}$$

Equation (1) is referenced through `cleveref`, so the word "Equation" is generated automatically.

3 Figure Examples [328 words]

Every figure below uses a placeholder file from `figures/`. The class sets `\graphicspath` to that folder, so `\includegraphics` needs only the file name.

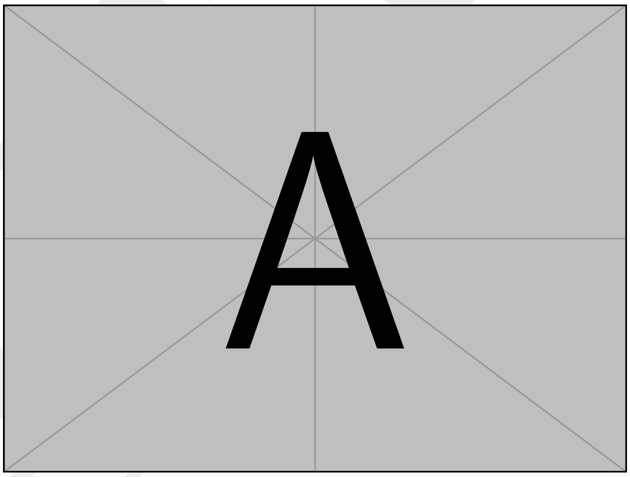


Figure 1: A single-column figure, sized with `width=\columnwidth`.

3.1 Single-column figure

This is the default case. Size the image with `width=\columnwidth` rather than a fixed length, so it adapts when the column model changes. The label goes after the caption, never before, or the reference number comes out wrong. See fig. 1.

3.2 Full-width figure

Use `figure*` to span both columns in the two-column model. It can only float to the top of a page, never to the middle or bottom, and never onto the page it was declared on. Declare it a page or two before you want it to appear. In the one-column model `figure*` behaves exactly like `figure`, so the source stays valid in both modes. See fig. 2.

3.3 Side-by-side subfigures

`subcaption` gives each panel its own letter and label. Reference the whole figure as fig. 3, or one panel as fig. 3a. Keep the two `subfigure` widths summing to slightly under 1, so the panels do not collide.

3.4 Stacked subfigures

Drop the `\hfill` and add a blank-line break to stack panels vertically instead.

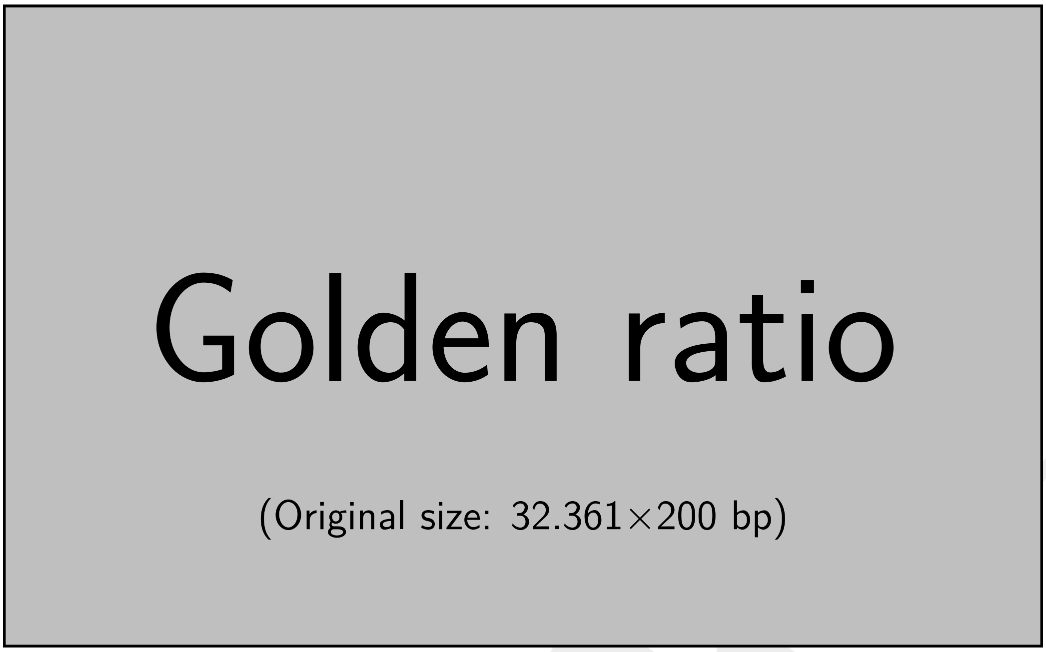
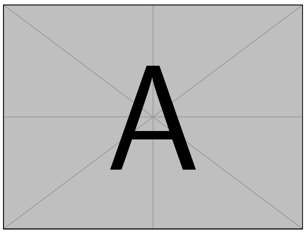
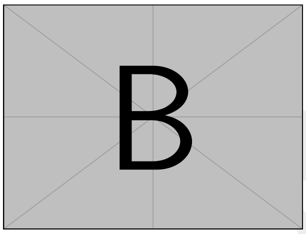


Figure 2: A full-width figure, declared with `figure*` and sized against `\textwidth`.



(a) First panel.



(b) Second panel.

Figure 3: Two subfigures side by side.

3.5 Text-wrapped figure

`wrapfig` lets prose flow around a small figure. It is fragile near page breaks, list environments, and section headings, so use it sparingly and check the output every time. Put it at the start of a paragraph, never inside one.

The paragraph that follows the `wrapfigure` environment is the one that wraps around it. Give the environment a width a little larger than the image itself, so the text does not press against the edge of the graphic. If the wrap breaks across a page, move the environment one paragraph earlier or later rather than fighting the placement options. Two or three sentences of surrounding prose are usually enough to fill the space beside a half-column image.

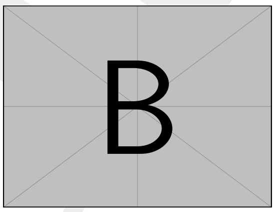
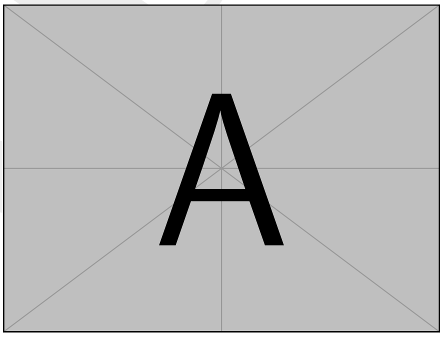


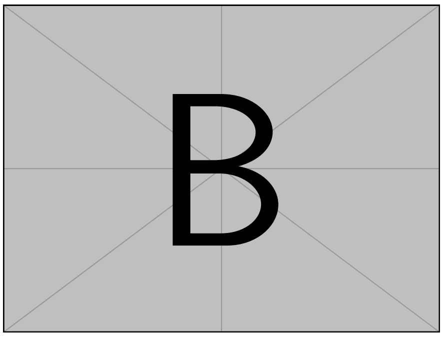
Figure 5: A wrapped figure.

3.6 Rotated and oversized figures

`adjustbox` handles an image that refuses to fit. `max width=\columnwidth` shrinks only when needed, and leaves smaller images untouched.



(a) Top panel.



(b) Bottom panel.

Figure 4: Two subfigures stacked vertically.

4 Table Examples [278 words]

All tables use `booktabs` rules. Never use vertical rules, and never use `\hline` where `\toprule`, `\midrule`, or `\bottomrule` will do. The caption sits above the table, unlike a figure caption, which sits below.

4.1 Simple table

The plain case, with a column specification of `l` for text and `r` for numbers. See table 2.

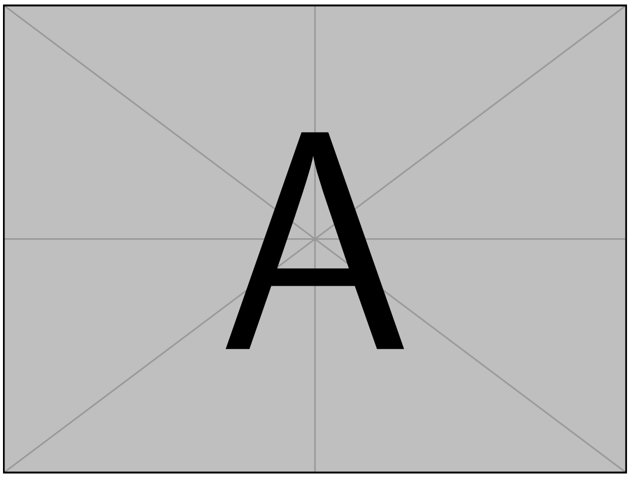


Figure 6: An oversized image, clamped by `adjustbox`.

Table 2: A simple table with `booktabs` rules.

Technical process	Papers	Share
System realisation	8281	96.02%
Verification	216	2.50%
Mission analysis	121	1.40%
Requirements definition	5	0.06%

Table 3: Grouped column headers with `\multicolumn` and merged rows with `\multirow`.

Phase	Process	Corpus	
		Count	Share
Concept	Mission analysis	121	1.40%
	Stakeholder needs	0	0.00%
Definition	Requirements	5	0.06%
	Architecture	402	4.66%

Table 4: A wrapping description column, using `tabularx` and the `L` column type.

Process	Description
Mission analysis	Establishes the problem space and the operational need that the system is meant to address.
Requirements	Transforms stakeholder needs into a verifiable specification of what the system shall do.
Verification	Confirms that the realised system meets the specification it was built against.

4.2 Grouped headers and merged cells

`\multicolumn` groups columns, and `\cmidrule(lr)` underlines only that group. `\multirow` merges cells downward. The `(lr)` trimming keeps adjacent rules from touching. See table 3.

4.3 Table with a wrapping text column

`tabularx` stretches one column to fill a target width. The class defines `L`, `C`, and `R` as left, centre, and right ragged versions of the stretching `X` column, which avoids the stretched justification that plain `X` produces. Set the target width to `\columnwidth` in the two-column model. See table 4.

4.4 Numeric alignment

`siunitx` aligns numbers on the decimal point through the `S` column type. Header text in an `S` column must be wrapped in braces, or the package will try to parse it as a number. Use `\num` and `\SI` in the body text too, so units and digit grouping stay consistent throughout the paper. See table 5.

4.5 Table with notes

`threeparttable` keeps footnotes attached to the table rather than to the page. The note markers are independent of the document footnote counter, so they do not disturb the numbering of ordinary footnotes. See table 6.

4.6 Full-width table

`table*` spans both columns, and carries the same top-of-page-only restriction as `figure*`. See table 7.

4.7 Oversized table

When a table is a little too wide, shrink it with `adjustbox` rather than reducing the font size by hand. If it is far too wide, the table is usually trying to do too much, and belongs in an appendix or split in two.

5 Other Elements [277 words]

5.1 Cross-references

Everything is referenced through `cleveref`, so the leading word is generated rather than typed. Write `\cref{tab:simple}` and get table 2, or `\Cref{fig:single}` at the start of a sentence and get Figure 1. Several references collapse automatically, so `\cref{tab:simple,tab:grouped,tab:wrapping}` gives tables 2 to 4. Never hard-code the word “Table” or “Figure” in front of a reference, because it defeats the whole mechanism.

Label prefixes keep the reference space legible. Use `fig:`, `tab:`, `eq:`, `sec:`, and `alg:`.

5.2 Citations

Citations are numeric and sorted, through `natbib`. Use `\citep` for a parenthetical citation, as in [1], and `\citet` for a textual one, as in Ishida [1]. Several keys in one command compress into a range, so `\citep{example2026,example2025}` gives [1, 2]. Page numbers go in the optional argument, as in [1, p. 14].

5.3 Equations

A numbered display uses `equation`, and stays inside the draft-mode line numbering.

$$I(A;B) = H(A) + H(B) - H(A,B) \tag{2}$$

Table 5: Decimal-aligned figures, using the `siunitx` S column.

Protocol	Rate (kbit/s)	Loss (dB)
BB84	1204.50	6.200
Decoy	980.75	11.480
MDI	12.05	24.000
TF	3.40	46.125

Table 6: A table carrying its own notes.

Source	Records[a]	Retained
IEEE Xplore	3412	3180
Scopus	6104	5444[b]

[a] Before de-duplication across the two databases.
[b] Excludes conference abstracts shorter than two pages.

Use `align` for several related lines, aligned on the relation symbol. Mark any line that does not need a number with `\nonumber`.

$$\begin{aligned} R &\geq q\left[1 - H_2(e_{\mathrm{b}}) - f H_2(e_{\mathrm{p}})\right] \nonumber \\ &\geq q\left[1 - 2H_2(e)\right] \end{aligned} \tag{3}$$

An unnumbered display uses `\[ ... \]`, and a case distinction uses `cases`.

$$\rho = \begin{cases} |0\rangle\langle 0| & \text{with probability } p, \\ |1\rangle\langle 1| & \text{otherwise.} \end{cases}$$

Equations (2) and (3) are referenced the same way as everything else.

5.4 Lists

`enumitem` controls the spacing. The `noitemsep` key removes the gaps between items, which matters in a two-column paper where vertical space is scarce.

- A tight bulleted item.
- A second one.

(i) A numbered item with a custom label format.
(ii) A second one.

Concept The phase in which the problem space is characterised.
Definition The phase in which requirements and architecture are settled.

5.5 Other fragments

A footnote looks like this.[1] Inline verbatim uses `\verb`, and a short quotation uses the `quote` environment.

> A displayed quotation, indented on both sides and set without quotation marks.

[1] Footnotes sit at the foot of the column in the two-column model.

6 Conclusion [22 words]

Replace this file with the conclusion. Keep the closing short, and state what the paper established rather than restating the structure.

Acknowledgment [9 words]

Replace with acknowledgements, or comment this section out.

References

[1] Hayato Ishida. Placeholder entry. *Journal of Placeholder Studies*, 1(1):1–10, 2026.
[2] Hayato Ishida. Second placeholder entry. In *Proceedings of the Placeholder Conference*, pages 100–110, 2025.

Table 7: A full-width table, declared with `table*`.

Process	Standard clause	Intent	Papers
Mission analysis	6.4.1	Characterise the problem space and the operational need.	121
Stakeholder needs	6.4.2	Elicit and agree the needs of those who will use the system.	0
Requirements	6.4.3	State verifiable system requirements derived from those needs.	5
Architecture	6.4.4	Define the structural options that could satisfy the requirements.	402

Table 8: A wide table clamped to the column with `adjustbox`.

Process	2021	2022	2023	2024	2025
Implementation	412	455	501	588	640
Integration	88	104	161	224	302
Verification	21	24	30	36	44

A Appendix [198 words]

Appendices usually read better in a single column, even when the body is set in two. Switching is a matter of one \onecolumn before the appendix material, as done in main.tex. In the one-column model the command is harmless, so the source stays valid in both modes.

A.1 Multi-page table

longtable breaks a table across pages, which an ordinary table float cannot do. It does not work inside a two-column page, which is the reason for the \onecolumn switch above. It is also not a float, so it appears exactly where it is written, and takes no placement argument.

The header rows are declared once and repeated automatically on every page.

Table 9: A table that breaks across pages, using longtable.

Process	Clause	Papers
Mission analysis	6.4.1	121
Stakeholder needs	6.4.2	0
Requirements definition	6.4.3	5
Architecture definition	6.4.4	402
Design definition	6.4.5	1044
System analysis	6.4.6	188
Implementation	6.4.7	5901
Integration	6.4.8	934
Verification	6.4.9	216

The four header and footer blocks are all optional. \endfirsthead runs on the first page only, \endhead on every later page, \endfoot at the foot of every page but the last, and \endlastfoot at the very end. Note the \\ after \label, which is easy to omit and produces a confusing error when it is.